

C2 Charge Carriers

Data: Magnitude of the charge on an electron = $1.60 \times 10^{-19} \text{ C}$

Free electron density of germanium [Ge] = 10^{20} m^{-3}

- C2.1 How many electrons are needed to carry a charge of -9 nC ?
- C2.2 An electron gun emits 3.2×10^{19} electrons in 15 s. What is the beam current?
- C2.3 A piece of germanium measures $2.0 \text{ mm} \times 1.0 \text{ mm} \times 2.0 \text{ cm}$. If the semiconductor is carrying a current of 20 mA between the two smallest faces, calculate the mean velocity of the free electrons.

CHAPTER C. ELECTRIC CIRCUITS

Answers

$$\text{C2.1 } \frac{9 \times 10^{-9}}{1.6 \times 10^{-19}} = 5.625 \times 10^{10}$$

$$\text{C2.2 } \frac{3.2 \times 10^{19} \times 1.6 \times 10^{-19}}{15} = 0.3413 \text{ A}$$

$$\text{C2.3 } \frac{0.02}{10^{20} \times 2 \times 10^{-6} \times 1.6 \times 10^{-19}} = 625 \text{ m s}^{-1}$$